STAFF REVIEW · H446 289 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A shared secret used to encrypt and decrypt.

2. Can be distributed for encryption or verification according to the scheme.

3. To protect its content from unintended readers.

4. $00000110_2 = 06_{16}$; high four bits masked away.

I Do — source worked steps

1. Predict first — The file contents or expected reference differ: the integrity check fails.

2. Show method — The mismatch does not by itself reveal whether alteration was accidental or malicious.

We Do — guided CHECK

1. Predict first
Different random salts make identical passwords produce different stored verifiers.
Show method
A costly password KDF slows guessing; salt is generally stored beside the verifier and need not be secret.

You Do — source mark schemes

1. Hash maps data to fixed-length digest [2], designed one-way [1]; encryption designed to recover plaintext with key [2]; hashes are not decrypted [1].

2. Compute digest of downloaded bytes [2]; compare with trusted expected digest [2]; match is strong integrity evidence [1]; compromised reference cannot establish authenticity [1].

3. Unique random salt per account [2], purpose-built slow KDF [2], prevents direct same-password verifier equality/precomputed attacks [1], raw or weak unsalted hashes expose credentials [1].

Exit Ticket — source answer

1. False. It computes a candidate verifier and compares it.